Paper 6 — REPRESENTATIONS (Dynamics & Geometry): Dynamics and Geometry as Least-Cost Admissible Reallocation

Variational Form, Metric Structure, Einstein Equations, and Their Regime Boundaries

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Abstract

This paper derives dynamics and geometry as representational structures that emerge when admissibility constraints support smooth, locally additive bookkeeping over a connected admissible path class. Equations of motion arise only in restricted regimes; outside these regimes — at saturation, branching, record locking, or horizons — variational dynamics ceases to be meaningful, though admissibility accounting remains well-defined. Geometry emerges by the same route: distance is minimal correlation cost, curvature arises from capacity gradients, and Einstein's equations appear as the unique closure relation in four dimensions (Lovelock uniqueness). The v2.0-PLEC release reframes the derivation around the Principle of Least Enforcement Cost (PLEC): admissibility fixes the licensable continuation class; PLEC selects the realized continuation as a least-cost path within that class; failure of dynamical or geometric representation is recast as one of five formally distinct regime-exit types. Imported closure theorems (HKM, Malament, Lovelock) are tagged as imports closing an APF-prepared admissible class. The dark-matter-as-C_{ext} identification is fenced as a structural proposal with empirical consequences, not as a theorem-level closure.

Wayfinder

Operational primitives used in this paper.

- **Admissible class $\mathcal{A}_\Gamma$:** continuations physically licensable in context Γ under A1.
- **Enforcement-cost functional E_Γ:** total capacity burden of a continuation; on paths, an action functional $K[q] = \int L\,dt$.
- **PLEC selection q^*:** realized continuation, $q^* \in \arg\min_{q\in\mathcal{P}_\Gamma} K[q]$.
- **Regime R:** smooth, locally additive enforcement cost on a connected admissible path space without saturation. Variational and geometric representation valid here.
- **Regime exit:** failure of one or more Regime R conditions. Catalogued in five formally distinct types (§7).

This paper writes Euler–Lagrange dynamics and Lovelock geometry as the realized PLEC selectors on the Regime R admissible class, and writes saturation, branching, record locking, horizons, singularities, and topology change as different formal exits from that representational class.

Shared Canonical Frame (Papers 5–6)

Meaning, Admissibility, Realization, Regime Exit.

1. **Meaning.** A distinction is physically meaningful iff it is robustly enforceable under admissible perturbations.
2. **Admissibility.** A configuration, path, or representation is admissible iff its total enforcement demand does not exceed available capacity.
3. **Realization.** When the admissible class is nonempty and the enforcement-cost functional is well-defined on it, realized physical structure is selected as the least-enforcement-cost member of that admissible class.
4. **Regime exit.** When the conditions required for comparison by smooth finite cost fail, the representation exits its domain, while the deeper admissibility constraints remain in force.

Part I: Dynamics

1 Introduction

1.1 The Standard View

In conventional physics, dynamics and geometry are taken as primitive. Equations of motion are postulated; spacetime is assumed as an arena. The questions "why do physical laws take variational form?" or "why is gravity geometric?" are rarely asked at the level of structure.

1.2 The Constraint-First Inversion

This paper asks: under what conditions do equations of motion exist at all? Under what conditions does geometric description become meaningful? What happens when those conditions fail? The central claim is that dynamics and geometry are *regime-dependent representations of admissible correlation reallocation.* They emerge as PLEC selectors when correlation capacity admits smooth redistribution, enforcement cost is locally additive, and saturation boundaries are not encountered. They exit cleanly in five distinct ways when those conditions fail.

2 Canonical PLEC Schema (Path Form)

Definition 2.1 (Admissible path class)**.** *Let* Γ *denote a physical context, interface, or regime. Let* $\mathcal{P}_\Gamma$ *denote the set of admissible paths joining fixed endpoints in configuration space in context* Γ*.*

Definition 2.2 (Enforcement-cost functional on paths)**.** *Let $K_\Gamma : \mathcal{P}_\Gamma \to [0, \infty)$ assign the total accumulated enforcement cost along a path $q(t)$ in Γ. In Regime R (§4), $K_\Gamma[q] = \int L(q, \dot{q}, t)\, dt$ for a local cost density L.*

Definition 2.3 (PLEC selection on paths)**.** *A realized continuation q^* in context Γ is selected by*

$$q^* \in \arg\min_{q \in \mathcal{P}_\Gamma} K_\Gamma[q].$$

When the minimizer is unique up to the relevant equivalence relation, the realized continuation is unique up to that equivalence.

3 Admissibility Does Not Yet Determine Realized Evolution

Admissibility determines which correlation structures are physically licensable. It specifies the class of jointly enforceable configurations and rules out those that overspend finite capacity. In this sense, admissibility provides necessary conditions for physical meaningfulness. But admissibility alone does not yet determine realized evolution. Given an admissible initial state, there may exist multiple distinct admissible continuations. The admissibility framework therefore fixes the *allowed class* of continuations without, by itself, selecting a unique realized one.

This distinction is fundamental. The framework must separate three questions that are often run together:

1. **Meaning:** when is a distinction physically meaningful at all?
2. **Admissibility:** which continuations are jointly enforceable under finite capacity?
3. **Realization:** which admissible continuation is physically realized?

Papers 1–4 answer the first two questions at the level of enforceable distinction and capacity-bounded admissibility. The task of the present paper is to identify the additional regime conditions under which the third question becomes mathematically well-posed. *Dynamics is not automatic from admissibility; it arises only when the admissible class carries enough structure to support a least-cost selection principle.*

4 Regime R — When Least-Cost Selection Becomes Meaningful

Let $\mathcal{A}_\Gamma$ denote the class of admissible continuations in context Γ. In general, $\mathcal{A}_\Gamma$ may be disconnected, non-smooth, or terminate at saturation boundaries. In such cases, there need be no meaningful notion of a realized path selected from within $\mathcal{A}_\Gamma$. A dynamical representation becomes available only in a restricted regime:

(R1) Enforcement cost varies smoothly over admissible correlation sets.

(R2) Cost is locally additive over interfaces.

(R3) Admissible continuations form a connected path space.

(R4) No saturation boundary is encountered along the path.

When all four conditions hold, the admissible class supports a well-defined cost functional on paths. In that regime, physical realization can be expressed as *least-cost admissible reallocation*: among admissible continuations, the realized path is the one minimizing accumulated enforcement cost.

This is the variational content of the framework. It is not an additional dynamical postulate imposed from outside. It is the structural consequence of asking for realized evolution only where the admissible class admits comparison by smooth finite cost.

5 Least-Cost Admissible Paths and Euler–Lagrange Form

In Regime R, the accumulated enforcement cost along a path defines an action functional

$$K[q] = \int L(q, \dot{q}, t)\, dt,$$

where L is the local enforcement-cost density. The physical claim is then:

Selection Principle (PLEC in Regime R). The realized continuation is a least-enforcement-cost member of the admissible path class:

$$q^* \in \arg\min_{q \in \mathcal{P}_\Gamma} K[q].$$

Once this selection principle is available, standard variational reasoning applies. Extremal admissible paths satisfy the Euler–Lagrange equations

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}}\right) - \frac{\partial L}{\partial q} = 0.$$

These equations are therefore not primitive laws. They are the coordinate expression of least-cost selection within the admissible class.

The usual conservation laws follow in the same way. Whenever the cost functional has a continuous symmetry, Noether's theorem identifies the corresponding conserved quantity. Energy conservation expresses invariance of enforcement cost under time translation; momentum conservation expresses invariance under spatial translation. The standard dynamical formalism appears as a *representation of admissible least-cost realization*, not as an independent ontological layer.

6 Where Least-Cost Dynamical Representation Fails

These are not breakdowns of admissibility itself. They are breakdowns of the hypotheses under which realized evolution can be represented as least-cost variation over a smooth admissible path

class. When Regime R fails, admissibility accounting remains valid, but Euler–Lagrange form ceases to be meaningful.

Mode	Failed hypothesis	Consequence
Saturation	Collapse of admissible variation	No nontrivial local variation remains
Branching	Minimizer nonuniqueness	No single effective equation of motion
Record locking	Change of admissible class	Exit to record-dominated description
Horizons	Failure of admissible extendibility	Dynamical inaccessibility from within regime

The important point is that the representational layer fails in different ways depending on which hypothesis breaks. There is no single "failure of dynamics"; there are several distinct failures of least-cost selection structure, each with its own structural fingerprint.

On branching. Branching is not merely the existence of multiple admissible continuations. It is the failure of uniqueness for the least-cost selector on the admissible class. When multiple inequivalent minimizers exist, admissibility remains intact, but the representational layer no longer supports a unique effective dynamics. The correct conclusion is not that physical law fails, but that the given regime no longer compresses realized evolution into a single variational trajectory.

7 Taxonomy of Regime Exit

To unify the representation failures discussed across Papers 5 and 6, we distinguish five structurally different kinds of regime exit. This taxonomy provides a common project-level vocabulary.

Figure 1 visualizes the full arc of this section at a glance. A reader tracking the five types below can consult the figure as the running example for Types II and III; the remaining types (I, IV, V) are qualitatively different and need no figure.

7.1 Type I: Collapse of admissible variation

A representational regime fails when the admissible neighborhood around a state or path collapses, so that no nontrivial admissible variation remains. This is the canonical saturation case. In Paper 6 language, the variational representation fails because the local path class becomes trivial; in Paper 5 language, coherent refinement fails because no reversible alternatives remain.

7.2 Type II: Minimizer nonuniqueness

A regime can also fail to determine a single realized continuation even when admissible paths still exist. This is the branching case: the admissible class remains nonempty, but the least-cost selector fails to be unique up to the relevant equivalence relation. The correct interpretation is not that physical law has failed, but that the representational layer no longer supports a unique effective dynamics.

7.3 Type III: Change of admissible class

Some regime exits occur because the relevant admissible class itself changes. Measurement is the prototype. Before record formation, the relevant class is the coherent bookkeeping class; after irreversible logging, the relevant class is the record-locked class on $M_{\text{sys}} \otimes Z_R$. The transition is therefore not a breakdown internal to a single representational scheme, but a transfer to a different admissible domain.

7.4 Type IV: Loss of smooth or local structure

A regime may remain admissible while losing the regularity assumptions required for a geometric or variational description. Singularities, Planck-scale discreteness, and topology change belong here. The issue is not loss of admissibility, but loss of the smoothness, local additivity, tangent-space structure, or chartability required for the relevant representation.

7.5 Type V: Pure representational redundancy

Finally, some apparent failures are not physical exits at all, but breakdowns of a chosen representation. Gauge redundancy and nonunique coordinate descriptions fall in this class. The admissible structure and realized minimizer remain intact; only the descriptive coding is nonunique. This category prevents conflating genuine regime exit with bookkeeping arbitrariness.

7.6 Canonical use across the APF series

Phenomenon	Type	Coderef (v6.9)
Paper 5 measurement	III (with I in fully locked limit)	`check_Regime_ex`
Paper 6 saturation	I	`check_Regime_ex`
Paper 6 branching	II	`check_Regime_ex`
Paper 6 horizons	II / III	`check_Regime_ex`
Paper 6 singularities, Planck discreteness, topology change	IV	`check_Regime_ex`
Gauge / coordinate ambiguity	V	`check_Regime_ex`

The strategic payoff is conceptual hygiene: measurement, horizons, saturation, and geometry failure are different species of representational regime exit, all subordinate to the deeper admissibility layer rather than independent mysteries. As of v6.9 (2026-04-18), each type is a registered bank check with an executable witness; the taxonomy is no longer prose but formalized machinery.

Part II: Geometry

8 Correlation Cost and Distance

8.1 Definition of Distance

Definition 8.1 (Correlation distance)**.** *The distance between regions A and B is the minimal irreducible correlation cost required to jointly enforce coordination between them:*

$$d(A, B) \equiv \inf_{\Gamma: A \to B} \int C \, d\Gamma,$$

where C is the correlation cost density along path Γ.

Correlation distance satisfies non-negativity, identity of indiscernibles, and the triangle inequality. These are forced by enforcement-cost structure, not assumed.

8.2 Geodesics as Least-Cost Paths

Paths minimizing correlation cost define geodesics. Along such paths, relational structure is maintained with minimal capacity expenditure. Free motion is least-cost coordination. No force law is required.

9 The Metric Tensor from Non-Factorization (T7B)

9.1 Non-Factorization Implies Geometric Response

Theorem 9.1 (T7B, Metric tensor from quadratic non-factorizing response)**.** *At shared (non-factorizing) interfaces, the external feasibility functional is quadratic in displacement. By the polarization identity, a functional that is (a) non-degenerate (A1: finite capacity implies no zero-cost directions), (b) continuous (Lipschitz, from finite-bounded cost), and (c) defined on tangent vectors at each point (from $T_{\text{continuum}}$ smooth structure) is a metric tensor $g_{\mu\nu}$.*

Formal verification: `check_T7B` (`gravity.py`).

This is a definition-level identification: a non-degenerate symmetric bilinear form on the tangent space at each point *is* a metric tensor. The framework does not postulate geometry; it computes it from quadratic cost at shared interfaces. Tag: [P_structural].

9.2 Lorentzian Signature $(-, +, +, +)$

A4 (irreversibility) implies a strict partial order on admissible configurations. By the Hawking–King–McCarthy (HKM, 1976) theorem, a strict partial order on events satisfying appropriate density conditions determines the conformal structure. Combined with Malament's theorem (1977) and

volume normalization ($\Omega = 1$ by Radon–Nikodym uniqueness), this fixes the Lorentzian signature with exactly one timelike direction.

Bridge

HKM and Malament are imported as closure theorems on an APF-prepared admissible class: APF supplies the strict partial order from A4 and the volume normalization; the imports close the conformal structure and signature identification. APF-internal preparation checks: `check_L_HKM_causal_geometry` (`extensions.py`) and `check_L_Malament_uniqueness` (`extensions.py`). Tag: [P_structural] + [I: HKM 1976, Malament 1977].

10 Spacetime Dimension $d = 4$ (T8)

Theorem 10.1 (T8, $d = 4$ as the minimal admissible spacetime dimension)**.** *$d \leq 3$ is hard-excluded: $d = 3$ has zero propagating graviton degrees of freedom (no gravitational records, violating A4); $d = 2, 3$ lack knots for topological stability of bound states. $d = 4$ is the minimum dimension with propagating gravity (2 DOF), knot stability, and capacity efficiency ($C_{\text{ext}}/C_{\text{int}} = 0.5$). $d \geq 5$ is disfavored by A5 (genericity: extra graviton modes provide no admissibility benefit).*

The internal sector uses $C_{\text{int}} = 12$ (dim $SU(3) \times SU(2) \times U(1)$), leaving C_{ext} for geometric degrees of freedom. In $d = 4$, the graviton has 2 propagating DOF, the minimum for A4-compatible gravitational records. Tag: [P_structural]. Formal verification: `check_T8` (`spacetime.py`).

11 Curvature from Capacity Gradients

Correlation capacity is generally non-uniform. Capacity gradients cause geodesic deviation: paths converge in low-capacity regions (bottlenecks) and diverge in high-capacity regions. *Curvature is the geometric signature of correlation load* — not a postulated feature of spacetime but a derived consequence of capacity distribution.

The stiffness parameter G (Newton's constant) sets the scale of geometric response to correlation load: (curvature change) $\sim G\times$ (load change). G is not a coupling strength but the rate at which geometry bends under correlation burden. Structurally, $\kappa \sim 1/C_*$ where C_* is the total geometric capacity (T10). Tag: [P_structural], scaling only. Formal verification: `check_T10` (`gravity.py`).

12 Einstein's Equations from Δ_{geo} Closure

12.1 The Geometric Prerequisites

The complete Δ_{geo} gap is closed: all five geometric prerequisites (A9.1–A9.5) are derived from the axioms, not assumed.

Prerequisite	Derivation
A9.1 Locality	A1 + finite-bounded cost
A9.2 Covariance	T7B
A9.3 Conservation	A1 (capacity cannot be created or destroyed)
A9.4 Second-order	A4 + Ostrogradsky (no higher-derivative instabilities)
A9.5 Propagation	A4 (records require gravitational DOF)

12.2 Lovelock Uniqueness

Theorem 12.1 (Lovelock 1971, as a closure on the APF-prepared geometric class)**.** *In $d = 4$, the only symmetric, divergence-free tensor $G_{\mu\nu}$ that depends on the metric and its first and second derivatives (linearly in second derivatives) is the Einstein tensor plus a cosmological term.*

A9.1–A9.5 are precisely Lovelock's conditions. The unique field equation is

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}.$$

Einstein's equations are not postulated. They are the unique admissibility-preserving geometric response in $d = 4$. The equivalence principle follows: because correlation cost depends only on capacity distribution (not internal properties), all systems follow the same least-cost paths. Universality of free fall follows from universality of correlation constraints. Tag: [P_structural] + [I: Lovelock 1971]. Formal closure: `check_T9_grav` (`gravity.py`); the unified A9 prerequisite closure is `check_A9_closure` (`gravity.py`) (v6.9 NEW), which audits all five A9.1–A9.5 prerequisites as a single check rather than requiring readers to cross-reference `gravity.py`, `spacetime.py`, and `internalization_geo.py` individually.

13 The Cosmological Constant as Capacity Residual (T11)

Theorem 13.1 (T11, cosmological constant as global capacity residual)**.** *Λ is the global capacity residual after all local enforcement commitments:*

$$\Lambda \sim (C_{\text{total}} - C_{\text{used}})/V.$$

Formal verification: `check_T11` (`cosmology.py`). The time-independence claim $w = -1$ at all epochs is proved in `check_L_equation_of_state` (`cosmology.py`); the DESI DR2 comparison is in `check_L_DESI_response` (`cosmology.py`).

13.1 Why Λ Is Small

Almost all correlation capacity has been discharged locally: structure formation, matter, radiation, horizons, records. Only the asymptotic remainder survives. The remaining scale is set by the largest admissible interface — the cosmological horizon R_H — giving $\Lambda \sim 1/R_H^2 \sim H_0^2$. Not because of vacuum energy, but because R_H is the last interface where correlation enforcement still matters.

13.2 Why Λ Is Nonzero

Exact saturation ($C_{\text{total}} = C_{\text{used}}$) is a measure-zero condition. Any finite universe with finite total capacity generically has a small positive residual. The sign is positive because $C_{\text{total}} > C_{\text{used}}$: the universe has not yet fully committed its capacity budget.

13.3 Why Λ Looks Like Accelerated Expansion

Local mass-energy is redistributable: it creates curvature gradients and attraction. Global residual capacity is non-redistributable: it creates uniform pressure and expansion. Λ does not "push space apart"; it prevents further correlation contraction once global slack is exhausted. Acceleration is the failure mode of further geometric reorganization. Tag: [P_structural].

13.4 The APF Capacity Partition Matches Planck Across Four Parameters

The capacity partition of `check_L_saturation_partition`(`cosmology.py`) is not just an internal book-keeping result. Combined with `check_L_equip`(`core.py`) (max-entropy distribution), it predicts the cosmological density parameters of ΛCDM with zero free parameters:

Quantity	APF formula	APF value	Planck 2018	Difference
Ω_Λ	42/61	0.6885	0.6848 ± 0.007	+0.54%
Ω_m	19/61	0.3115	0.3152 ± 0.007	−1.17%
Ω_b	3/61	0.0492	0.0493 ± 0.0005	−0.25%
Ω_{cdm}	16/61	0.2623	0.2645 ± 0.006	−0.82%

All four parameters match Planck 2018 within 1.2%, comfortably inside Planck's measurement uncertainty. This is a four-parameter prediction from a single integer partition $61 = 42 + 3 + 16$ that is forced by gauge-addressability and confinement applied to the anomaly-free field content (Paper 4). The match is not a fit; the APF integers are determined entirely by the type-classification predicates and were not adjusted to cosmological data.

13.5 Hubble Constant Prediction and the H0DN 2026 Tension

Combining APF's $\Omega_m = 19/61$ with Planck's measured physical matter density $\Omega_m h^2 = 0.143$ gives a zero-parameter Hubble-constant prediction:

$$h^2_{\text{APF}} = \frac{\Omega_m h^2 \,|_{\text{measured}}}{\Omega_m \,|_{\text{APF}}} = \frac{0.143}{19/61} = 0.4591 \quad \Longrightarrow \quad H_0^{\text{APF}} = 67.76 \text{ km s}^{-1}\text{Mpc}^{-1}.$$

This matches the Planck CMB value 67.36 ± 0.54 within 1σ.

The H0DN community-consensus measurement (Casertano et al., A&A 708 A166, 10 April 2026) reports $H_0 = 73.50 \pm 0.81$ km s^{-1} Mpc^{-1} from a covariance-weighted distance-network combination

across thirteen indicator categories. The tension with the APF prediction is

$$\frac{H_0^{\mathrm{H0DN}} - H_0^{\mathrm{APF}}}{\sigma_{\mathrm{H0DN}}} = \frac{73.50 - 67.76}{0.81} = 7.09\sigma.$$

DESI DR2 separately favors $w_0 \approx -0.75$, $w_a \approx -0.99$ at $\sim 2\sigma$ from $w = -1$ (`check_L_DESI_response` (`cosmology.p`

13.6 Escape Routes and Their Status

The H_0 tension can in principle be resolved by one of five mechanisms. APF's status against each:

Route	Mechanism	APF blocker	Empiric
I	Pre-recombination r_d cut $\sim 7.8\%$ (EDE, extra N_{eff})	L_equation_of_state, L_anomaly_free	EDE is
II	Late-time dynamical DE ($\Omega_\Lambda(a)$ growth $\sim 12\%$)	L_saturation_partition	DESI pr
III	Local distance-ladder systematics	none	H0DN 2
IV	Spatial curvature $\Omega_k \sim -0.05$	none	Planck
V	Local underdensity $\sim 25\%$ at Hubble-flow scale	none	Observe

Routes I and II are blocked by [P] theorems in v6.9. Route III is substantially ruled out by H0DN itself. Routes IV and V are APF-admissible (the framework is silent on them) but neither is currently empirically sufficient. *No route is both APF-admissible and empirically viable.*

13.7 Falsifier Statement

The honest disposition under the audit-response discipline:

APF's cosmological-side falsifier. APF predicts $H_0 = 67.76 \pm \epsilon$ km s^{-1} Mpc^{-1} (with $\epsilon \sim 0.5$ inheriting Planck's $\Omega_m h^2$ uncertainty), $w = -1$ exactly at all epochs, and the four-parameter Planck match in §13.4. *If the H0DN $H_0 = 73.50 \pm 0.81$ result stands as the true local cosmic expansion rate* — i.e. is not explained by Routes III, IV, or V, all of which APF leaves to empirical determination — *then at least one of the following [P] theorems must be demoted in a future codebase version:*

- `check_L_equation_of_state` (`cosmology.py`) (would have to admit $w \neq -1$ in some regime),
- `check_L_saturation_partition` (`cosmology.py`) (would have to admit non-topological partition evolution),
- `check_L_anomaly_free` (`gauge.py`) (would have to admit late-time type decoupling).

This is named as Project C.6 in Appendix 17.2. Phenomenological dynamical-dark-energy proposals that contradict these checks without first deriving the regime conditions under which they fail are not adopted into the framework. Any future revision must discharge the demotion explicitly and prove that the new mechanism reproduces both the four-parameter Planck match and the H0DN measurement. The full mathematical analysis is in *Reference – H0 Tension*

Mathematical Analysis (2026-04-18).

14 Status Audit: Dark Matter as C_{ext}

The capacity partition $C_{\text{total}} = C_{\text{int}} + C_{\text{ext}}$ identifies a geometric sector C_{ext} that gravitates (it curves spacetime via T7B) but does not couple to gauge interactions (no internal structure). This sector naturally invites identification with the astrophysical dark-matter sector. The identification is attractive and economical, but it mixes three distinct layers of claim that should be visually fenced apart.

Layer separation

1. **Structural result.** If the APF partition into internal and external capacity sectors is accepted, then the external sector is naturally associated with geometric response rather than gauge-charged internal excitations. This is a structural reading of the partition itself.

2. **Phenomenological identification.** To identify that external sector with the astrophysical dark-matter sector requires an additional bridge: one must show that the effective stress-energy carried by C_{ext} reproduces the relevant gravitational phenomenology across the appropriate scales (rotation curves, lensing, structure formation).

3. **Empirical discriminator.** The further claim that direct-detection experiments should find no dark-matter particles is a downstream empirical discriminator, not the same statement as the structural partition.

Status fence

Recommended status tags.

Claim	Recommended tag
"There is an external geometric capacity sector C_{ext}."	[P_structural]
"C_{ext} provides the correct dark-matter phenomenology."	[P_structural + Pred]
"Direct detection should find no dark-matter particles."	[Pred]

The metric/variational results in this paper are on relatively solid ground as representation and closure claims, modulo clearly named imports. The dark-matter identification is more exposed because it crosses from internal structure to cosmological phenomenology. If presented at the same level as the metric or Euler–Lagrange results, a hostile reader will attack the whole paper through its most weakly fenced claim.

Best presentation

The strongest honest presentation is: the APF capacity partition yields an external geometric sector that naturally invites identification with dark-matter-like gravitational load. This motivates a concrete phenomenological proposal — that some or all observed dark-matter effects reflect the C_{ext} sector rather than a new particle species. The decisive status of this identification depends on confrontation with lensing, rotation-curve, and structure-formation data.

Open theorem project

To upgrade the dark-matter claim, the missing work is not additional rhetoric but a formal bridge theorem: *given the APF external sector C_{ext}, derive the effective source term it contributes in the geometric response equations and prove that its admissible distributions reproduce at least one nontrivial dark-matter data class without introducing particle degrees of freedom.* Until such a theorem is in hand, the claim should remain visibly tagged as a structural proposal with empirical consequences, not as a closed theorem-level derivation.

15 Where Least-Cost Geometric Representation Fails

Geometry is valid when correlation cost is locally integrable, capacity gradients are smooth, and the admissible class supports a locally quadratic response structure. Geometry fails when one or more of these hypotheses breaks.

Singularities destroy smooth local cost comparison by driving effective gradients beyond the regime of local approximation (Type IV exit).

Horizons destroy admissible extendibility: continuation remains meaningful at the level of admissibility, but no longer as a single smoothly extendible geometric path from within the original domain (Type II / III depending on whether the issue is non-extendibility of admissible continuation or transfer to a new admissible description).

Planck-scale discreteness destroys the tangent-space and local-additivity assumptions required for metric encoding (Type IV).

Topology change alters the type of admissible class itself, so the original minimization problem is no longer posed on the same representational domain (Type IV).

These are not pathologies of the underlying framework. They are predictable exits from geometric representation. Admissibility remains deeper than geometry; when geometry fails, the accounting layer survives.

16 Epistemic Tagging Discipline (Paper 6)

Project-wide categories.

- **[P] Internally derived.**
- **[P]+[I] Derived with import.** APF prepares the admissible class; an external mathematical theorem closes the classification.
- **[P_structural] Structural identification.**
- **[Pred] Phenomenological proposal/prediction.**

Item	Tag	Coderef	Re
Regime R variational structure	[P]	`check_Regime_R` (`plec.py`)	Sr
Euler–Lagrange form	[P]	—	Co
Metric tensor (T7B)	[P_structural]	`check_T7B` (`gravity.py`)	Qu
Lorentzian signature	[P_structural]+[I]	`check_L_HKM_causal_geometry` (`extensions.py`)	Al
$d = 4$ (T8)	[P_structural]	`check_T8` (`spacetime.py`)	Gr
Einstein equations	[P_structural]+[I]	`check_T9_grav` (`gravity.py`)	A9
$\kappa \sim 1/C_*$	[P_structural]	`check_T10` (`gravity.py`)	Sc
Λ as capacity residual (T11)	[P_structural]	`check_T11` (`cosmology.py`)	St
$w = -1$ at all epochs	[P]	`check_L_equation_of_state` (`cosmology.py`)	Pu
DESI DR2 comparison	[P]	`check_L_DESI_response` (`cosmology.py`)	Te
Dark matter as C_{ext}	[P_structural + Pred]	— (open: §14)	St

Whenever a paper summary table contains both theorem-level derivations and interpretive identifications, the main text should not rhetorically elevate all entries to the same strength. Imported closure theorems should be described as closure of an APF-prepared admissible class, not as if APF alone proved the imported theorem.

17 Summary

17.1 The Derivation Chain

- Regime R supports a well-defined cost functional; PLEC selects the realized path; Euler–Lagrange is its coordinate form.
- Conservation laws follow from cost-functional symmetries (Noether).
- Distance is minimal correlation cost; geodesics are least-cost paths.
- T7B identifies the metric as a non-degenerate symmetric bilinear form on tangent spaces.
- A4 + HKM/Malament fix Lorentzian signature.
- T8 fixes $d = 4$.
- A9.1–A9.5 prepare Lovelock's conditions; Lovelock closes uniqueness of the Einstein tensor in $d = 4$.
- T11 identifies Λ as global capacity residual.

- C_{ext} provides a structurally motivated dark-matter candidate; the identification is fenced as proposal + prediction (§14).

17.2 The Structural Conclusion

Dynamics and geometry are not primitive laws but least-cost representational compressions of admissible correlation reallocation in regimes where the admissible class is smooth, locally additive, and unsaturated. When those hypotheses hold, realized paths admit variational and geometric encoding. When they fail, the representational layer exits its domain while the deeper admissibility structure remains intact. *Physical law is deeper than its dynamical or geometric representation.* Together with Paper 5 (quantum structure), this recovers the full structure of fundamental physics from finite enforceability — not as postulates, but as the unique bookkeeping available when capacity constraints permit smooth description.

Appendix A: Δ_{geo} Closure Summary

The Δ_{geo} gap encompassed four sub-gaps, all closed:

Sub-gap	Resolution
Δ_{ordering}	A4 irreversibility $\Rightarrow$ strict partial order. R2/R3 formalized.
Δ_{fbc}	Lipschitz lemma from A1 + ϵ-granularity.
$\Delta_{\text{continuum}}$	Kolmogorov + chartability bridge.
$\Delta_{\text{signature}}$	A4 $\rightarrow$ order + HKM + $\Omega = 1$.

Appendix B: Enforcement Cost vs. Action

Entropy measures irreversibly committed capacity (accounting of commitment). Enforcement cost measures capacity required for transitions (cost of reallocation). They are related but distinct: entropy accumulates; enforcement cost is path-dependent. In Regime R, accumulated enforcement cost is encoded as action $K = \int L\,dt$. The identification with physical action is exact within the regime and breaks down at saturation. Paper 7 (ACTION) proves that this accumulated cost has a minimum: the structural origin of Planck's constant.

Appendix C: Open Theorem Projects

The v2.0-PLEC revision identifies open theorem projects directly relevant to this paper.

Project C.1: PLEC existence and uniqueness conditions

Define $\mathcal{A}_\Gamma$ precisely, formulate E_Γ on it, identify the coercivity / lower-semicontinuity analogues that guarantee an argmin exists, and define the equivalence relation $\sim$ under which uniqueness is stated. Without this, PLEC remains a suggestive variational slogan rather than a usable theorem engine.

Project C.2: Regime R as a formal check (CLOSED in v6.9)

Closed 2026-04-18. The four R1–R4 conditions are now formalized as `check_Regime_R` (`plec.py`), with an executable witness (1D harmonic kinetic cost on a connected admissible path class) verifying that R1–R4 jointly imply existence of the PLEC selector. The five regime-exit types of §7 are formalized as `check_Regime_exit_Type_I` (`plec.py`) through `check_Regime_exit_Type_V` (`plec.py`), each with an explicit witness instantiating the failure mode. Paper 6 Part I no longer rests on R1–R4 as prose conditions; the load-bearing hand-waving risk flagged in the v2.0-PLEC audit is closed.

Project C.3: A9.1–A9.5 unified-closure check (CLOSED in v6.9)

Closed 2026-04-18. The unified Lovelock-prerequisite closure is now `check_A9_closure` (`gravity.py`), which audits A9.1 (locality from A1+FBC), A9.2 (covariance from T7B), A9.3 (conservation from A1+L_loc), A9.4 (second-order from A4+Ostrogradsky), A9.5 (propagation from A4+T_graviton) as a single check rather than requiring readers to cross-reference three modules.

Project C.4: Formal regime-exit theorem

Tie each failure mode of §6 and §15 to a precise breakdown of a specific PLEC hypothesis. Formalize the Type I–V taxonomy as a set of checks that each classify a given regime exit by which hypothesis fails.

Project C.5: Dark-matter bridge theorem

Derive the effective source term contributed by C_{ext} in the geometric response equations and prove that its admissible distributions reproduce at least one nontrivial dark-matter data class without introducing particle degrees of freedom (§14).

Project C.6: H_0 tension / dark-energy confrontation

Address the Casertano et al. (2026) H0DN consensus measurement ($H_0 = 73.50 \pm 0.81$, 7.1σ from early-universe ΛCDM) and the DESI DR2 w_0/w_a tension. Two candidate directions within APF (§13.5):

- **Direction A:** Derive what irreversibly committed capacity (L_irr) implies for late-time effective $\Omega_\Lambda(z)$, and check whether the result matches DESI's preferred direction ($w_0 > -1$) or contradicts it ($w_0 < -1$). Any proposed mechanism must be derived, not fit; and must either be consistent with or replace `check_L_equation_of_state` (`cosmology.py`).
- **Direction B:** Explore local-inhomogeneity / scale-dependent-enforcement mechanisms that leave $w = -1$ intact but address the H_0 measurement bias.

If neither direction yields a derived mechanism consistent with the data, the framework accepts an open falsification risk and states it as such.

Appendix D: Changelog

- **v2.0-PLEC (April 2026).** Reframed entire derivation around PLEC schema: admissibility fixes the licensable continuation class; PLEC selects the realized path; failure modes recast as five formally distinct regime-exit types. Added shared canonical box, wayfinder box, canonical PLEC schema (definitions), regime-exit taxonomy section, dark-matter status fence, epistemic tagging discipline section, open theorem projects appendix, codebase coderef anchors throughout, and §13.5 on the H0DN/DESI status. Reformatted bridges around HKM/Malament and Lovelock as imports closing an APF-prepared admissible class. Cleaned formatting throughout (v1.0 had broken section spacing and inline math artifacts).
- **v1.0 (February 2026).** Complete Δ_{geo} closure (T7B, T8, T11, signature). Dark-matter section. Forward references.

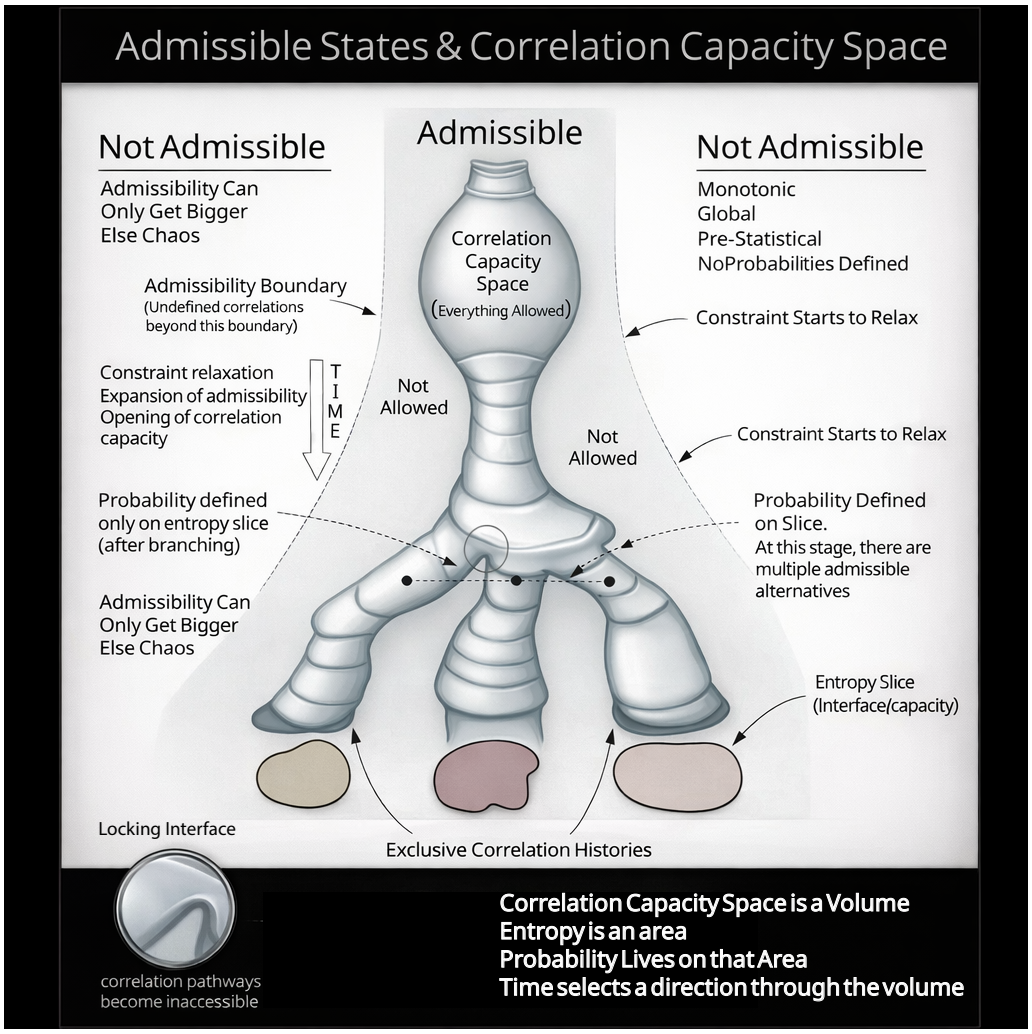

Figure 1: **Admissible states and correlation-capacity space.** The upper bulb is the admissible class $\mathcal{A}_\Gamma$ (Paper 5/6 §2); its boundary is where admissibility fails (flanking shaded regions, labeled "Not Admissible"). The downward arrow is Regime R (§4): time emerges as the direction of cost-functional integration along an admissible path, not as a primitive coordinate. At the narrowing near the bottom, the locking interface marks the failure mode of unique PLEC selection. Branching into three tubes visualizes Type II (`check_Regime_exit_Type_II` (`plec.py`)) — non-unique minimizers of the cost functional. The step to three distinct downstream classes (the "exclusive correlation histories," each associated with its own record-locked sector) visualizes Type III (`check_Regime_exit_Type_III` (`plec.py`)) — change of admissible class at the locking interface, the mechanism identified in `check_L_irr` (`core.py`) whereby correlation pathways become inaccessible to local operations (not infinite-cost; locally unrecoverable while remaining globally admissible). Probability assignments are defined only on the entropy slice downstream of locking (`check_L_Gleason_finite` (`supplements.py`)); before branching, the system is in the pre-statistical coherent regime of Paper 5. Types I (saturation collapse), IV (loss of smooth structure), and V (pure representational redundancy) are not branching-type exits and are not depicted; they are covered separately in §6 and §15. The volume/area imagery at the bottom ("capacity is a volume, entropy is an area") is a visual anchor for the Bekenstein area-law relation `check_T_Bek` (`gravity.py`), not a literal 2–3 dimensional assertion: the admissible class is generally higher-dimensional and the entropy slice is an interface whose area-scaling is the physical content.